Where airlines lose time: carrier-attributable delay in US domestic flights

This analysis uses US DOT (BTS) flight data rather than Singapore data, since CAAS/data.gov.sg only publishes aggregate monthly figures for Changi, and no flight-level, cause-coded delay data is publicly available. BTS was chosen as the best available public dataset with the granularity needed for this kind of analysis.

Question. Which airports or carriers show disproportionately high carrier-attributable delay, and does time-of-day or day-of-week make it worse?

What "carrier-attributable" means. BTS splits each delay of 15+ minutes into five causes. *Carrier* delay covers causes inside the airline's control: maintenance, crew, cleaning, fuelling, baggage. BTS has no separate "mechanical" field, so this is **not** a measure of mechanical problems. Weather, air-traffic-system (NAS) and security delays sit outside the airline's control. Late-arriving-aircraft delay is knock-on delay from an earlier flight and is kept separate.

Data: Jan 2024 – Dec 2025, 13,844,730 completed flights. Carrier-attributable delay averages 5.2 min per flight and is 33% of all cause-coded delay minutes.

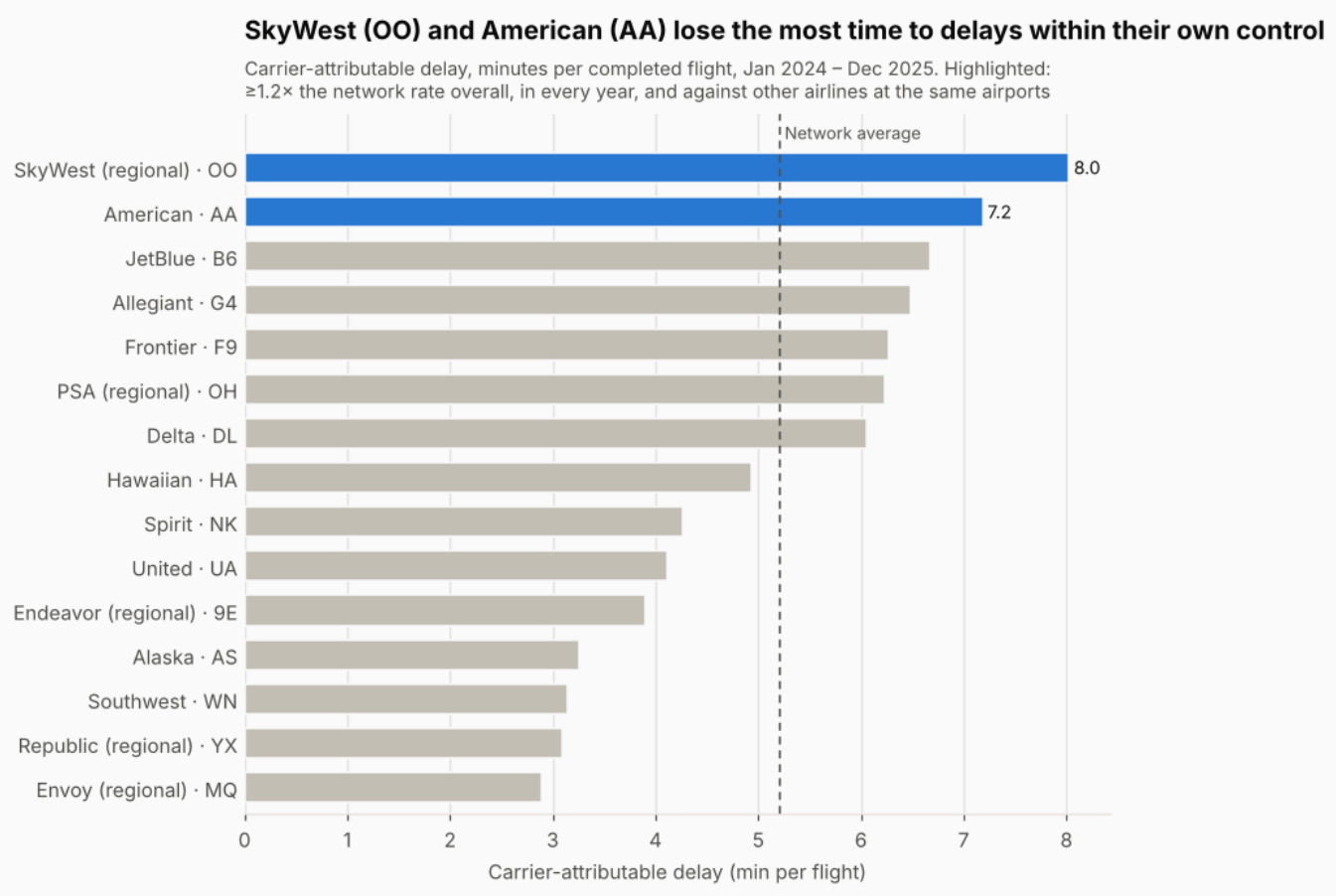
Summary

SkyWest (OO) and American (AA) have disproportionately high carrier-attributable delay: 8.0 and 7.2 minutes per flight, against a network average of 5.2. They stay at least 1.3× the rate of other airlines flying from the same airports, in 2024 and 2025 separately.

Time of day makes it worse. Across the network, carrier-attributable delay per flight is 2.0× higher at 22:00 than at 10:00, and Sunday is the worst day of the week. But for both flagged carriers the biggest excess is in the **first wave**: departures scheduled before 09:00.

Recommendation: SkyWest (OO) and American (AA) should target first-wave departure readiness. Bringing their pre-09:00 departures down to the network's pre-09:00 rate would remove about **20,600 carrier-attributable delay hours a year**. That is before counting the knock-on delay those early flights pass to the rest of the day.

1 · Carriers



Finding 1 · Carriers. Two airlines meet every test:

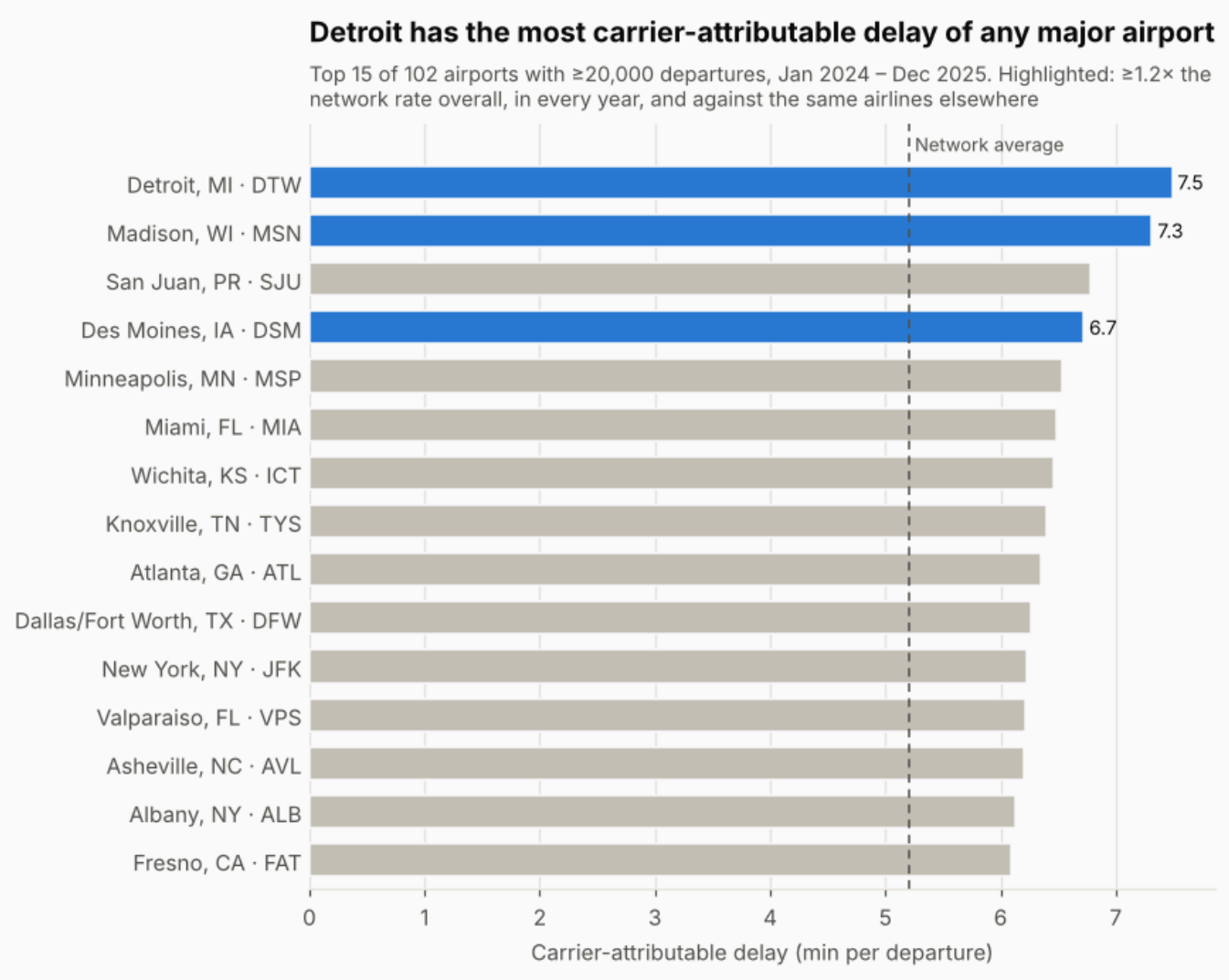
- **SkyWest (OO)**: 8.0 min per flight, 1.54× the network; 1.34× after controlling for airports flown; 2024: 1.51×, 2025: 1.56×. 36,500 excess delay hours a year.
- **American (AA)**: 7.2 min per flight, 1.38× the network; 1.33× after controlling for airports flown; 2024: 1.43×, 2025: 1.32×. 31,600 excess delay hours a year.

They fail in different ways. When a carrier-caused delay happens, the typical one lasts 35 min at SkyWest and 17 min at American (network: 19 min). They also happen on 11% and 15% of flights (network: 12%). So SkyWest's delays are longer, and American's are more frequent.

Above 1.2× overall but not in every year or after the mix control: JetBlue (B6), Allegiant (G4) and Frontier (F9). For example, JetBlue's rate fell from 1.54× in 2024 to 1.02× in 2025.

2 · Airports

Grouped by **origin** airport, where the delay was incurred.



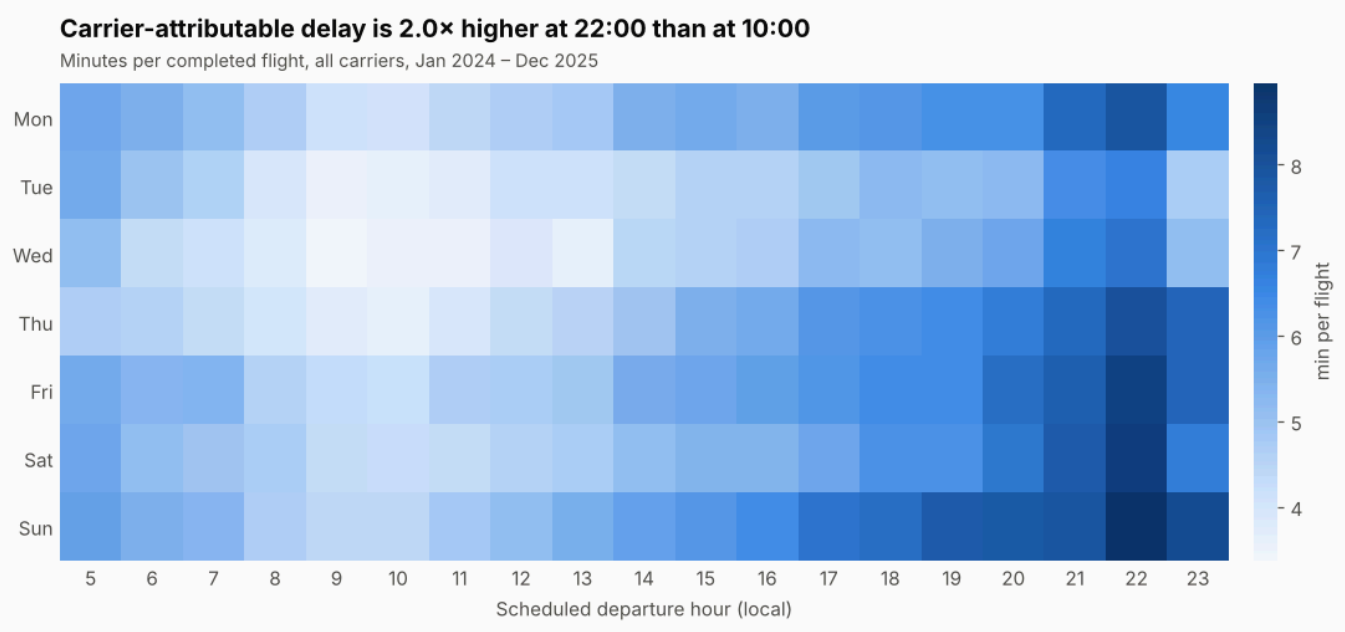
Finding 2 · Airports. 3 of 102 major airports meet every test:

- **Detroit, MI (DTW)**: 7.5 min per departure (1.44×; 1.25× against the same airlines elsewhere). 250,604 departures; main operators Delta (DL) 47%, SkyWest (OO) 21%.
- **Madison, WI (MSN)**: 7.3 min per departure (1.40×; 1.26× against the same airlines elsewhere). 24,110 departures; main operators SkyWest (OO) 21%, American (AA) 17%.
- **Des Moines, IA (DSM)**: 6.7 min per departure (1.29×; 1.27× against the same airlines elsewhere). 29,443 departures; main operators American (AA) 15%, United (UA) 14%.

Airports matter less than carriers. The airport effect is smaller once airline mix is controlled for, and the flagged airports add up to 5,600 excess hours a year, against 68,100 for the flagged carriers.

3 · Time of day and day of week

By **scheduled** departure hour (local time), so a late flight doesn't move into a later hour.



Finding 3 · Time of day and day of week. Across the network, carrier-attributable delay is lowest in the late morning (3.9 min per flight at 10:00) and climbs through the afternoon and evening to 8.0 min at 22:00. Sunday (6.0 min) is the worst day and Wednesday (4.5) the best. The weekday effect is smaller than the hour-of-day effect.

For the flagged carriers, the gap over the network is widest early in the day, when the network itself is at its best. Before 09:00 they run 1.76× and 1.51× the network rate. That band is 21% and 26% of their flights but 28% and 33% of their excess delay hours. Most of these flights start the day from an overnight position. That points at readiness (maintenance sign-off, crew, aircraft preparation), though the BTS data can't say which.

Each carrier's rate against the network in each time band, and the excess delay hours a year:

Scheduled departure	SkyWest (OO)	American (AA)
<09:00	1.76× · 10,300 h/yr	1.51× · 10,300 h/yr
09–13	1.62× · 8,700 h/yr	1.33× · 4,900 h/yr
13–17	1.48× · 8,100 h/yr	1.29× · 5,100 h/yr
17–21	1.46× · 8,400 h/yr	1.33× · 7,100 h/yr
21+	1.36× · 1,700 h/yr	1.40× · 3,600 h/yr

Recommendation

Fix the first wave at SkyWest (OO) and American (AA). Their pre-09:00 departures run 1.8× and 1.5× the network's carrier-attributable delay rate. Bringing them to the network rate would remove about **20,600 delay hours a year** (SkyWest (OO) 10,300, American (AA) 10,300). Likely places to look, which BTS can't separate: overnight maintenance sign-off, crew show-up and aircraft readiness (cleaning, catering, fuelling) at the stations that launch the first wave. All of these are inside the airline's control, which is what this metric measures.

The number is conservative. A late first departure pushes delay onto later flights, and BTS books that as *late-arriving aircraft* delay (40% of all delay minutes), which is not counted here.

Caveats

- **Carrier delay ≠ mechanical delay.** BTS records maintenance in the same field as crew, cleaning, fuelling and baggage.
- **Short delays aren't counted.** Delay causes are only recorded for flights 15+ minutes late.
- **Knock-on delay is left out.** Late-arriving-aircraft delay is partly the carrier's own knock-on delay, but BTS doesn't record what caused it. Leaving it out of "carrier-attributable" makes the estimate conservative.
- **SkyWest is a regional carrier.** It flies under the United, Delta, American and Alaska brands but operates and reports these flights itself. Its delays are its own operation's, not the brand's.
- **The CrowdStrike outage doesn't change the main finding.** The IT outage of 19 July 2024 hit Delta hardest. With 19–28 July 2024 removed, SkyWest and American stay flagged (see appendix). Frontier, just under the bar in 2024 (1.17×), also crosses it (1.22×), because the outage week pushed the network average up.
- **Delay causes are self-reported** by airlines. BTS covers US domestic flights by carriers above a revenue threshold, so the results may not carry over to other markets such as Changi.

Future work (out of scope for this pass)

- Weather delay by airport and season
- Route and distance effects
- Station-level first-wave analysis for the flagged carriers (which origin airports drive the pre-09:00 excess)
- Predicting carrier-attributable delay; an interactive dashboard (Streamlit)

Appendix: supporting tables

Cause breakdown, network-wide

	cause	share	hours_per_year
0	Late-arriving aircraft	0.40	715,305.44
1	Carrier	0.33	601,247.00
2	NAS (air traffic system)	0.20	367,547.69
3	Weather	0.06	110,871.55
4	Security	0.00	2,772.75

Carriers (`index` = against the network; `adj_index` = against other airlines at the same airports; `index_YYYY` = that year only; `excess_hours` = whole period)

carrier	flights	carrier_min_per_flight	carrier_delayed_rate	carrier_share	on_time_rate	index
SkyWest (regional) · OO	1559088	8.02	0.11	0.49	0.80	1.54
American · AA	1918957	7.18	0.15	0.33	0.74	1.38
JetBlue · B6	462386	6.68	0.17	0.33	0.74	1.28
Allegiant · G4	244774	6.49	0.11	0.32	0.77	1.24
Frontier · F9	397672	6.27	0.13	0.27	0.72	1.20
PSA (regional) · OH	460355	6.23	0.12	0.29	0.75	1.19
Delta · DL	2010253	6.05	0.10	0.45	0.81	1.16
Hawaiian · HA	156968	4.93	0.14	0.60	0.84	0.95
Spirit · NK	446824	4.27	0.11	0.26	0.77	0.82
United · UA	1532195	4.12	0.11	0.27	0.79	0.79
Endeavor (regional) · 9E	195072	3.90	0.07	0.31	0.84	0.75
Alaska · AS	482250	3.26	0.08	0.28	0.77	0.62
Southwest · WN	2781434	3.14	0.12	0.27	0.79	0.60
Republic (regional) · YX	629350	3.09	0.07	0.24	0.82	0.59
Envoy (regional) · MQ	567152	2.89	0.09	0.22	0.79	0.56

Carriers with the CrowdStrike outage window (19–28 Jul 2024) removed

carrier	flights	carrier_min_per_flight	index	adj_index
SkyWest (regional) · OO	1538381	7.98	1.57	1.35
American · AA	1891248	7.07	1.39	1.34
JetBlue · B6	455913	6.64	1.30	1.28
Allegiant · G4	240650	6.43	1.26	1.09
Frontier · F9	391598	6.26	1.23	1.24
PSA (regional) · OH	454190	6.12	1.20	1.12
Delta · DL	1985709	5.68	1.11	1.05
Hawaiian · HA	154775	4.93	0.97	1.48
Spirit · NK	439342	4.12	0.81	0.82
United · UA	1512564	4.00	0.78	0.80
Endeavor (regional) · 9E	190341	3.66	0.72	0.64
Alaska · AS	474648	3.25	0.64	0.76
Southwest · WN	2740003	3.11	0.61	0.72
Republic (regional) · YX	621444	3.03	0.59	0.61
Envoy (regional) · MQ	558675	2.86	0.56	0.50

Airports, top 20

origin	flights	carrier_min_per_flight	carrier_delayed_rate	carrier_share	on_time_rate	ind
Detroit, MI · DTW	250604	7.49	0.13	0.50	0.80	1.4
Madison, WI · MSN	24110	7.32	0.09	0.40	0.80	1.4
San Juan, PR · SJU	70036	6.78	0.14	0.37	0.77	1.3
Des Moines, IA · DSM	29443	6.72	0.09	0.38	0.80	1.2
Minneapolis, MN · MSP	237065	6.54	0.11	0.48	0.80	1.2
Miami, FL · MIA	217835	6.48	0.15	0.32	0.73	1.2
Wichita, KS · ICT	20995	6.46	0.10	0.37	0.80	1.2
Knoxville, TN · TYS	28936	6.40	0.10	0.35	0.79	1.2
Atlanta, GA · ATL	646726	6.35	0.14	0.44	0.78	1.2
Dallas/Fort Worth, TX · DFW	613057	6.26	0.17	0.31	0.72	1.2
New York, NY · JFK	223141	6.23	0.12	0.37	0.77	1.
Valparaiso, FL · VPS	20261	6.21	0.09	0.36	0.81	1.
Asheville, NC · AVL	21692	6.20	0.12	0.31	0.76	1.
Albany, NY · ALB	24169	6.13	0.10	0.35	0.79	1.
Fresno, CA · FAT	24883	6.09	0.10	0.44	0.82	1.
Houston, TX · IAH	228651	6.09	0.13	0.33	0.76	1.
Milwaukee, WI · MKE	55393	6.08	0.09	0.39	0.81	1.
Grand Rapids, MI · GRR	36210	6.02	0.10	0.36	0.79	1.
Colorado Springs, CO	25624	5.94	0.10	0.38	0.80	1.

origin	flights	carrier_min_per_flight	carrier_delayed_rate	carrier_share	on_time_rate	ind
· COS						
Philadelphia, PA · PHL	193921	5.86	0.14	0.30	0.76	1.

By scheduled departure hour and by weekday

dep_hour	flights	carrier_min_per_flight	carrier_delayed_rate	index
0	23179	6.75	0.10	1.30
1	7394	6.78	0.10	1.30
2	2810	4.85	0.09	0.93
3	1801	6.55	0.10	1.26
4	824	7.47	0.10	1.43
5	392090	5.49	0.06	1.05
6	962364	5.04	0.06	0.97
7	988059	4.84	0.07	0.93
8	954934	4.34	0.08	0.83
9	781277	3.96	0.08	0.76
10	860819	3.95	0.09	0.76
11	868224	4.19	0.10	0.80
12	821886	4.48	0.11	0.86
13	827952	4.61	0.12	0.88
14	797477	5.11	0.13	0.98
15	796118	5.36	0.14	1.03
16	796908	5.45	0.15	1.05
17	867902	5.89	0.15	1.13
18	835674	6.10	0.15	1.17
19	753475	6.28	0.16	1.21
20	615005	6.62	0.17	1.27
21	459680	7.32	0.16	1.40
22	315814	8.02	0.16	1.54
23	113064	6.69	0.13	1.28

day_of_week	flights	carrier_min_per_flight	carrier_delayed_rate	index
Mon	2088250	5.39	0.12	1.03
Tue	1893059	4.55	0.09	0.87
Wed	1910400	4.46	0.10	0.86
Thu	2046133	5.09	0.12	0.98
Fri	2061923	5.57	0.12	1.07
Sat	1788896	5.32	0.12	1.02
Sun	2056069	6.00	0.13	1.15

Flagged carriers by time band

carrier	band	flights	min_per_flight	network_min_per_flight	vs_network	excess_hours_per_ye
OO	**<09:00**	332825	8.55	4.85	1.76	10,266.
	09–13	403161	6.73	4.15	1.62	8,679.
	13–17	396623	7.57	5.13	1.48	8,078.
	17–21	350358	9.06	6.19	1.46	8,388.
	21+	76121	10.16	7.49	1.36	1,692.
AA	**<09:00**	505932	7.31	4.85	1.51	10,349.
	09–13	428369	5.51	4.15	1.33	4,856.
	13–17	420656	6.59	5.13	1.29	5,133.
	17–21	418669	8.21	6.19	1.33	7,065.
	21+	145331	10.46	7.49	1.40	3,603.